

Data Project - Literature Review

Ori Sinvani - 325770824

Harel Brener - 214179012

Nehoray Chalfon - 325833531

Project Advisor - Ofer Zevin

Abstract

This project addresses the growing challenge of detecting illicit activity in cryptocurrency networks, where the pseudonymous nature of blockchain transactions is increasingly exploited for money laundering, terror financing, fraud, and other financial crimes. Although blockchain data is publicly available, its massive scale, complex structure, and dynamic behavior make manual analysis and traditional rule-based compliance tools insufficient. The target users of this system include financial institutions such as banks and cryptocurrency exchanges that must meet strict regulatory requirements, security and law-enforcement agencies monitoring illicit funding flows, and businesses or individuals seeking to assess the risk of interacting with a specific wallet address.

The project proposes the design and evaluation of an automated risk assessment system that leverages graph-based machine learning to analyze blockchain transaction data. Since blockchain activity is naturally represented as a graph of wallets and transactions, the system focuses on Graph Neural Network (GNN) approaches as its methodological foundation. In particular, scalable and inductive models such as GraphSAGE are surveyed due to their ability to operate on large, evolving graphs and to generalize to previously unseen wallet addresses—a critical requirement in real-world blockchain environments. Unlike existing solutions that rely primarily on static blacklists or heuristic rules, the proposed approach aims to learn structural and behavioral patterns directly from the transaction graph, enabling proactive detection of suspicious activity rather than purely reactive flagging.

The unique value of this project lies in its emphasis on structural graph patterns, temporal behavior, and generalization across datasets and threat types.

By surveying advanced GNN architectures, temporal extensions, and imbalance-aware learning techniques, the project explores whether illicit activity exhibits consistent structural signatures that persist across different crime categories and time periods. This perspective moves beyond narrow, dataset-specific models and aims to support earlier, more robust, and more generalizable detection of illicit behavior. As a result, the proposed system has the potential to enhance compliance workflows, support faster data-driven decision making, and improve the effectiveness of blockchain-based financial crime prevention compared to existing approaches.

General Project Description

Client and Organizational Context

While the system is being developed in an academic setting, it is designed to address critical compliance needs for specific external stakeholders in the financial and security sectors. The primary target audience for this solution includes financial institutions, such as banks and cryptocurrency exchanges, which face stringent regulatory requirements and must approve or reject customer wallets. Additionally, the system is designed for security agencies tracking illicit funding sources involved in terror financing or fraud, as well as businesses and private individuals seeking to verify the safety of a counterparty's wallet before executing a transaction.

Existing Workflows

In the current operational landscape, the detection of illicit cryptocurrency activity is predominantly manual or semi-automated. Compliance officers typically rely on copying a wallet address and searching for it in static databases or blocklists to check for known flags. These workflows rely heavily on rule-based filtering, where a wallet is flagged only if it has been explicitly blacklisted previously. This approach is reactive rather than proactive. It fails to detect new malicious actors who create fresh wallets that, while not yet blacklisted, are structurally connected to criminal networks. The massive volume of transactions and the complexity of the blockchain graph overwhelm human analysts using these traditional tools.

The Business Problem and Proposed Solution

The core business challenge is the tension between blockchain anonymity and financial regulation. Criminal actors exploit the pseudonymity of cryptocurrencies, specifically Bitcoin, to launder money, utilizing complex webs of transactions to obfuscate the origin of funds.

To address this, we are building an automated Risk Assessment System designed to bridge the gap between complex blockchain data and actionable compliance decisions. The system consists of two main components: a Web Interface and a Predictive Backend.

The workflow is straightforward: a compliance officer inputs a specific Wallet ID into the dashboard. The backend then acts as the system's "brain," employing a deep learning model to analyze the wallet's transaction history and its connections to other entities on the blockchain. The system processes this graph data in real-time and outputs a clear Risk Classification, potentially accompanied by a probability score. This allows financial institutions to make immediate, data-driven decisions to freeze funds or reject transactions before financial crimes are finalized.

Survey of Relevant Technologies

Technologies and algorithms considered for the project

The focus is on graph-based learning approaches, as blockchain data is naturally represented as a graph of transactions and wallet addresses. In particular, the project considers Graph Neural Networks (GNNs), including Graph Convolutional Networks (GCN) and GraphSAGE, as core modeling approaches. These models have been widely studied in prior academic work on blockchain anti-money-laundering and serve as strong baselines and scalable alternatives, respectively.

In addition, the survey includes topology-adaptive GCN-style layers that allow information to propagate across multiple hop distances, as well as attention-based and residual GNN variants that improve robustness to noisy or heterogeneous transaction neighborhoods. Since blockchain activity evolves over time, temporal modeling approaches—such as combining GNN embeddings with recurrent sequence models (LSTM or GRU)—are also reviewed as potential extensions. Finally, the project considers class-imbalance handling techniques and traditional machine-learning models (e.g., Random Forest and Logistic Regression) primarily as baselines and comparison points rather than as the final intended solution.

Advantages, disadvantages, and maturity of the surveyed technologies

Graph Convolutional Networks are among the earliest and most established graph learning architectures. Their primary advantage is conceptual simplicity and strong alignment with existing academic literature on illicit transaction detection. However, standard GCNs do not scale well to very large graphs due to full-batch training requirements and may suffer from over-smoothing when deeper architectures are used. Despite these limitations, GCNs are considered highly mature and remain a common reference model in blockchain AML research.

GraphSAGE was developed to address scalability and inductive learning limitations of earlier GNNs. By sampling local neighborhoods, it enables efficient training on large and evolving graphs and allows inference on previously unseen nodes, such as newly created wallet addresses. The main disadvantages of GraphSAGE include sensitivity to sampling strategies and potential loss of global structural information. Nonetheless, GraphSAGE is widely adopted in both research and industry and is considered a mature alternative to transductive GNN models.

Topology-adaptive GCN-style layers extend the expressiveness of standard GCNs by incorporating information from multiple hop distances in a single layer. This capability is particularly relevant in blockchain analysis, where illicit behavior often involves multi-step transaction chains. These models are less mature than basic GCNs and introduce additional complexity and risk of overfitting, but they represent an important research direction for capturing richer transaction patterns.

Temporal modeling approaches, such as recurrent neural networks applied to graph embeddings, aim to capture the evolution of transaction behavior over time. While they offer the potential to detect recurring laundering cycles and temporal anomalies, they significantly increase computational cost and model complexity. As a result, temporal GNN approaches are still an active area of research rather than a standardized solution.

Attention-based and residual GNN variants further enhance model flexibility by learning the relative importance of neighboring nodes and stabilizing deeper architectures. These models often outperform simpler GNNs in noisy or heterogeneous graphs but require more careful tuning and computational resources. Their maturity lies between classical GCNs and more experimental temporal GNN architectures.

Finally, class-imbalance handling techniques are essential due to the rarity of illicit activity in blockchain datasets. Although such techniques are well established in traditional fraud

detection, their application in graph-based learning remains an evolving research area, particularly when preserving graph structure is critical.

Motivation for considering these technologies

The technologies surveyed in this project were selected to provide a comprehensive understanding of the design space for blockchain illicit-activity detection. Simpler and well-established models such as GCN serve as reference points for evaluating more advanced approaches, while scalable and inductive models such as GraphSAGE address practical deployment considerations. More expressive architectures, including topology-adaptive, attention-based, and temporal models, are reviewed to assess their potential to capture complex transaction patterns that simpler models may miss. Additionally, imbalance-aware learning techniques are considered indispensable for any realistic solution in this domain, regardless of the final model choice.

Literature Review

The field of cryptocurrency forensics has evolved rapidly from simple heuristic-based methods to sophisticated Deep Learning approaches. As blockchain data is inherently structured as a network of transactions, Graph Neural Networks (GNNs) have emerged as the dominant methodology for detecting illicit activity. This review synthesizes papers in the field, relevant to our work, focusing on Graph Convolutional Networks (GCNs), temporal modeling, and the specific challenges of class imbalance in Bitcoin transaction graphs.

Graph Neural Networks for Fraud Detection

The foundational approach in the field focuses on classifying individual transactions or wallets in isolation. **Asiri et al. (2025)** demonstrated that optimized Graph Convolutional Networks (GCNs) can achieve high accuracy (98.5%) on the Elliptic dataset. Their work validates our choice of GCN as a baseline, showing that structural features combined with cost-sensitive training outperform traditional tabular classifiers. Building on this, **Zhang et al. (2024)** introduced GAT-ResNet, utilizing Graph Attention Networks (GAT). They addressed a key limitation of standard GCNs—handling noisy data—by using attention mechanisms to "weigh" the importance of neighbors. This confirms that advanced GNN layers are necessary to filter out irrelevant connections in noisy blockchain data.

Structural and Subgraph Analysis

While node-level models effectively identify simple illicit actors, they often fail to detect sophisticated criminals who utilize long chains of intermediary wallets to obfuscate their tracks. To address this gap, recent research has shifted towards analyzing larger structural patterns. **Ouyang et al. (2024)** proposed subgraph representation learning, arguing that money laundering schemes, such as "peel chains," manifest as distinct geometric motifs within the graph rather than isolated bad nodes. Their findings indicate that analyzing the subgraph (the broader neighborhood structure) captures complex operations that simple node-centric models might miss. This validates our project's approach of analyzing deeper multi-hop connections rather than just immediate neighbors.

Addressing Real-World Challenges: Time and Imbalance

Two major hurdles in operationalizing these models are the temporal nature of blockchain data and the scarcity of illicit labels. **Alarab & Pragoonwit (2022)** tackled this with a temporal graph-based LSTM. They integrated standard GCNs with LSTM networks to model how transaction behavior evolves over time, addressing the "cold start" problem via active learning. Similarly, the RL-GNN Fusion (2025) framework addressed the challenge of real-time detection in imbalanced streams. By combining GNNs with Reinforcement Learning, they proposed a way to dynamically adjust fraud thresholds, offering a method to handle "concept drift" where criminals change their tactics to evade static models.

Data Foundations and Benchmarking

Finally, high-quality data is the prerequisite for all these models. While the Elliptic dataset is the standard benchmark used in the studies above, The Real-CATS Dataset (2023) provides a critical expansion into modern threat landscapes (e.g., terrorism financing). This paper serves as the foundation for our multi-dataset strategy, ensuring our system is trained on diverse, modern criminal patterns rather than just historical Bitcoin thefts.

Relevance to the Current Project

While the reviewed literature establishes GNNs as a powerful tool for blockchain forensics, a significant gap remains in the generalizability of these models across diverse threat landscapes. Most existing studies (e.g., **Asiri et al.**, **Zhang et al.**) train and test on single-source datasets—primarily the Elliptic dataset—effectively optimizing their models for specific, historical crime patterns like 2016-era Bitcoin theft. They do not address whether these structural signatures hold true for modern, distinct threats such as terrorism financing or cybercrime found in newer datasets like Real-CATS.

Our Core Methodological Innovation:

Cross-Dataset Generalization: We are moving beyond single-dataset benchmarks to build a Unified Risk Model. By training on the Elliptic++ transaction graph (focused on theft/Ponzi schemes) and validating on the Real-CATS wallet graph (focused on modern cybercrime and extortion), we are testing the hypothesis that illicit activity carries a "universal structural signature." This approach challenges the prevailing method of building specialized, narrow models and aims to prove that graph embeddings can be robust across different time periods and crime types.

Multi-Entity Robustness: Current academic models are often "specialists"—good at detecting one type of crime (e.g., Ponzi schemes). By integrating two distinct datasets with non-overlapping threat categories (Theft vs. Cybercrime/Extortion), our project aims to create a "generalist" system. This addresses the critical blind spot in current literature, where models fail when facing new types of financial crime that do not resemble their training data.

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EDA

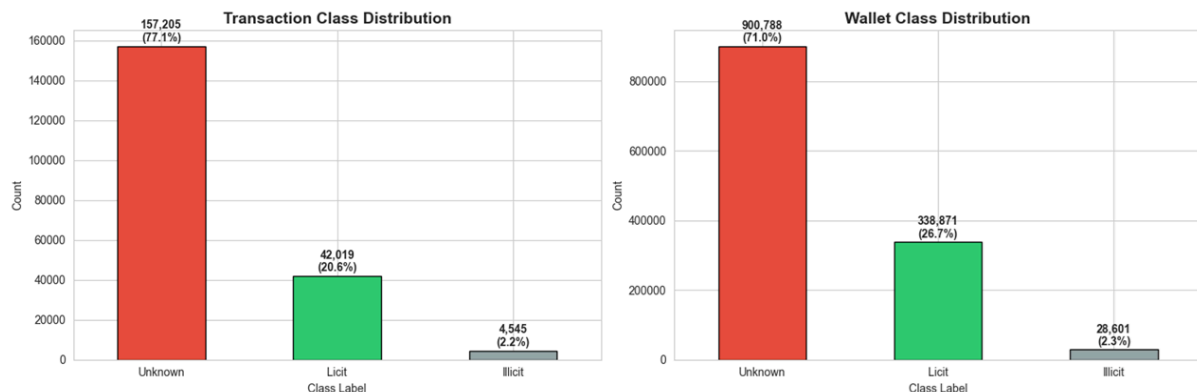
We have 2 datasets we would like to use in this project:

Elipctic++:

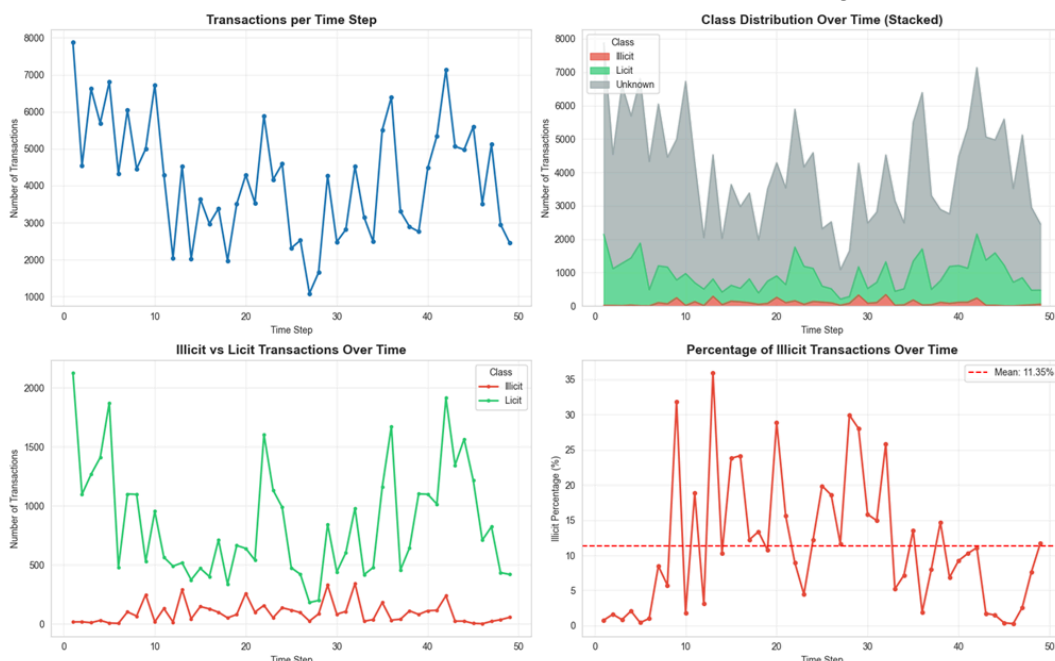
Dataset Shapes:

- Transactions: (203769, 186)
- Wallets: (1268260, 59)
- Transaction edges: (234355, 2)

Most of this dataset's transactions features are anonymous, and represented as local_feature/ aggregated_feature.



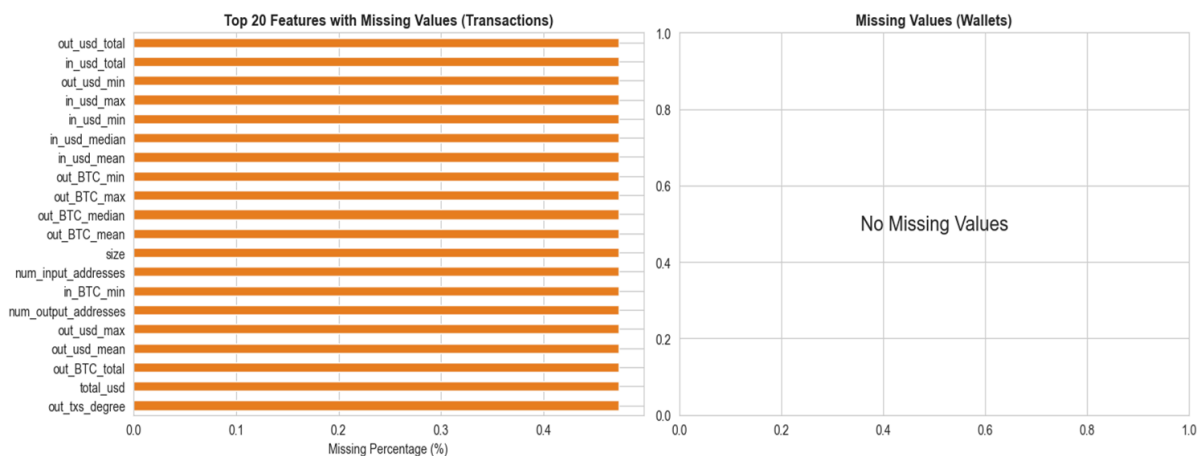
This figure shows the class distribution for transactions and wallets in the Elliptic++ dataset. In both cases, the majority of entities are labeled unknown, while licit entities form a smaller portion and illicit entities make up only a very small fraction. This highlights a **strong class imbalance**, which is an important consideration for model training and evaluation.



A time step is a fixed time window that groups together transactions that happened around the same time, allowing the dataset to track how activity changes over time.

This figure summarizes the temporal dynamics of transactions in the Elliptic++ dataset across discrete time steps:

- Top-left: The total number of transactions per time step shows strong temporal variability, indicating non-stationary activity in the network.
- Top-right: The stacked class distribution highlights that unknown transactions dominate throughout time, while licit transactions form a smaller but stable portion and illicit transactions remain relatively rare.
- Bottom-left: A direct comparison of licit vs. illicit transactions confirms the severe class imbalance, with illicit transactions consistently much fewer than licit ones.
- Bottom-right: The percentage of illicit transactions fluctuates over time, with spikes at certain time steps, but averages around ~11%, suggesting periods of heightened illicit activity rather than a constant rate.



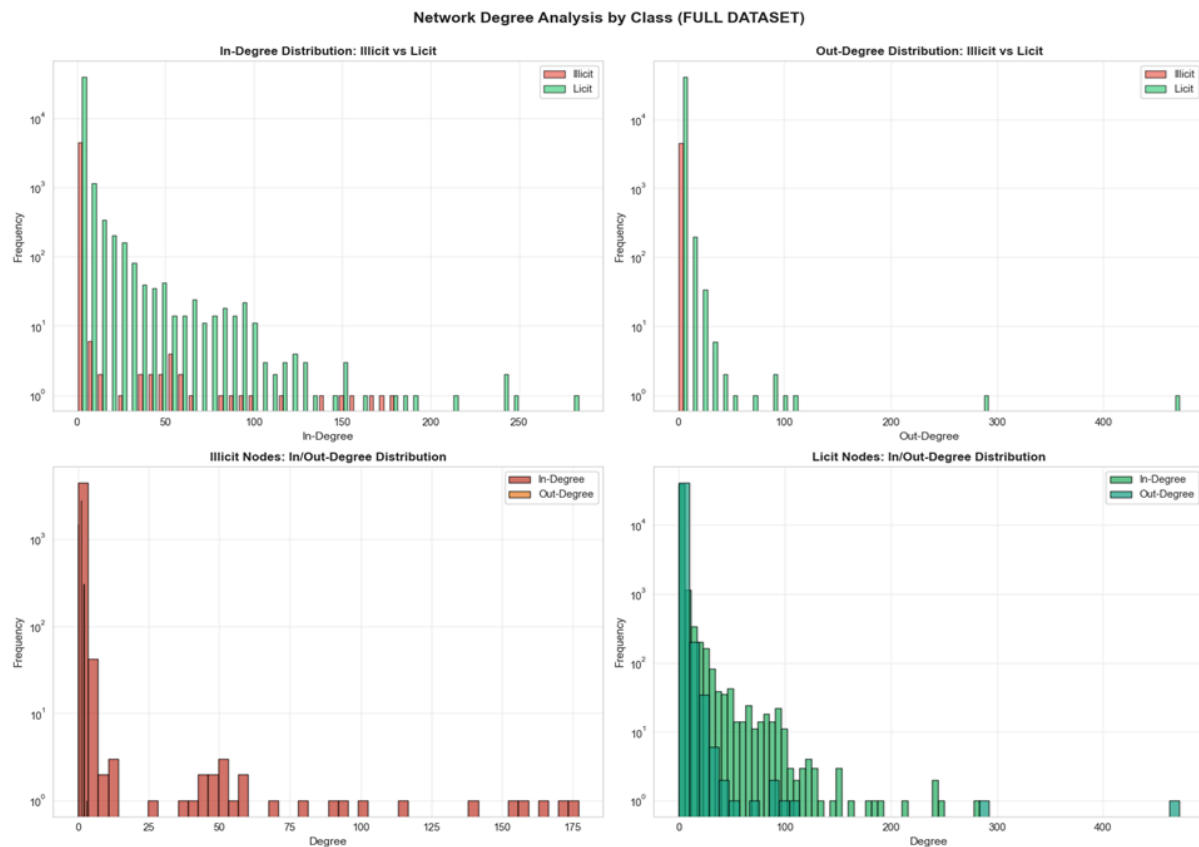
This figure examines missing values in the Elliptic++ dataset at both the transaction and wallet levels:

- Left panel: The top 20 transaction-level features with missing values all exhibit a very low and uniform missingness (~0.45–0.5%). These features are primarily aggregate statistics (e.g., USD/BTC totals, min/max/mean, address counts), suggesting that **missing values likely arise from edge cases** such as transactions without valid inputs/outputs rather than systematic data quality issues.
- Right panel: Wallet-level features contain no missing values, indicating a clean and fully observed representation at the wallet aggregation level.



This figure compares class-conditional feature distributions after normalizing for class size, allowing trends to be observed independently of class imbalance:

- Left (total_usd): Illicit transactions are more concentrated at lower transaction values, with a sharper peak, while licit transactions exhibit a broader distribution and heavier right tail, indicating higher-value and more variable transfers.
- Right (size): Illicit transactions are strongly skewed toward smaller transaction sizes, whereas licit transactions span a wider range and include larger, more complex transactions.



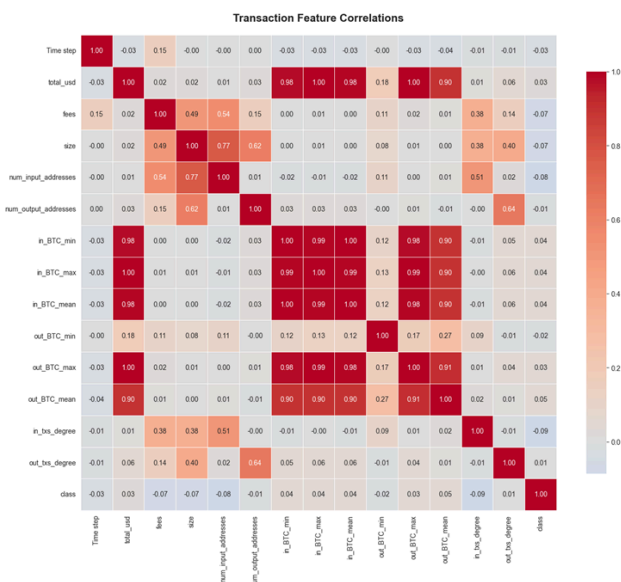
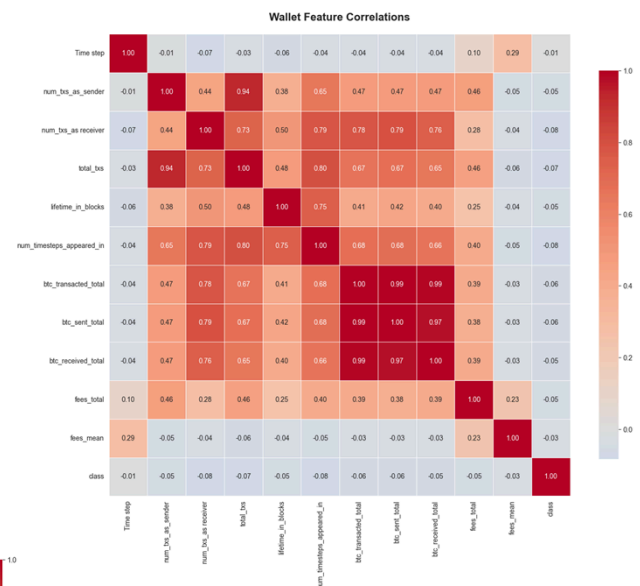
This figure analyzes network degree properties of nodes in the Elliptic++ transaction graph, comparing illicit and licit behavior across the full dataset:

- Top row: Both in-degree and out-degree distributions are highly right-skewed, but licit nodes exhibit much heavier tails, reaching significantly higher degrees. This indicates that **licit entities tend to participate in many more transactions** and act as hubs in the network.
- Bottom-left (Illicit nodes): Illicit nodes are mostly low-degree, with the majority having very small in- and out-degrees, and only a few reaching moderate connectivity.
- Bottom-right (Licit nodes): Licit nodes show a broad degree spectrum, including high-degree nodes consistent with exchanges, services, or aggregation points.

From	To	Count
Unknown	Unknown	131,778
Unknown	Licit	45,386
Licit	Licit	33,930
Licit	Unknown	15,116
Unknown	Illicit	3,993
Illicit	Unknown	1,458
Illicit	Illicit	998
Illicit	Licit	915
Licit	Illicit	781

This table shows how transactions move between unknown, licit, and illicit nodes in the Elliptic++ dataset. **Most transactions happen between unknown nodes**, since many entities are not labeled. Transactions involving licit nodes are much more common than those involving illicit ones. **Only a small number of transactions occur between illicit nodes**, meaning illicit activity rarely forms tightly connected groups. Instead, illicit nodes usually interact with unknown or licit nodes, blending into the wider transaction network.

This heatmap shows correlations between wallet-level features in the Elliptic++ dataset. Most activity-related features (number of transactions, total BTC sent/received, lifetime, and timesteps active) are strongly positively correlated, indicating they all capture overall wallet activity. In contrast, fee-related features and the class label show weak correlations with most variables, suggesting they provide more independent information.



This heatmap shows correlations between transaction-level features in the Elliptic++ dataset. Value-related features (USD and BTC totals, min/max/mean) are very strongly correlated, indicating they describe the same underlying transaction magnitude. Structural features such as transaction size and number of input/output addresses are moderately

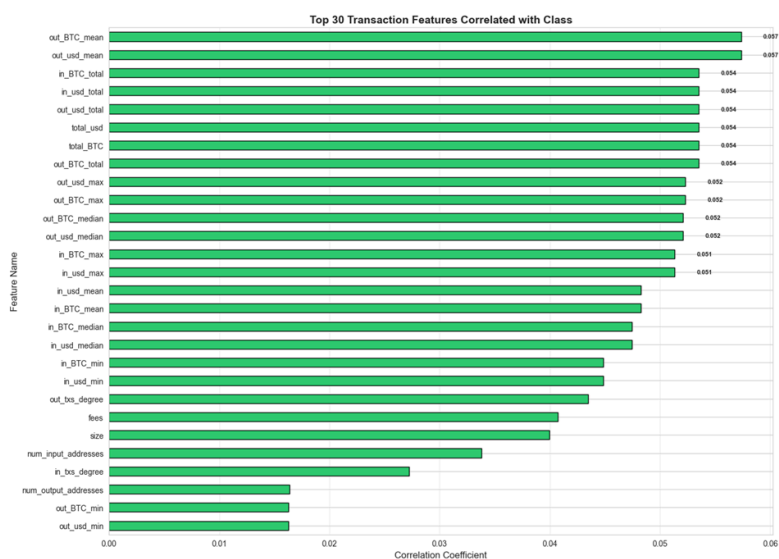
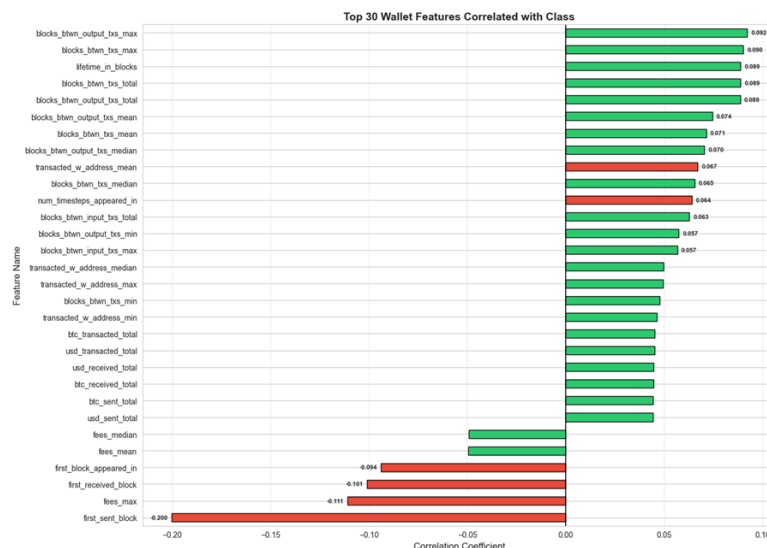
correlated with each other, capturing transaction complexity. The class label and time step show weak correlations with most features, suggesting that transaction characteristics, rather than time alone, drive class differences.

This bar chart shows the wallet-level features most correlated with the class label in the Elliptic++ dataset.

Features related to temporal behavior and network position (blocks between transactions, lifetime in blocks, and timesteps active) have the strongest positive correlations, suggesting that

how long and how regularly a wallet is active matters more than raw volume. In contrast, features tied to early activity

timing and fee statistics show negative correlations, indicating that wallets appearing earlier or with certain fee patterns are more likely to be associated with illicit behavior.



This bar chart shows the transaction-level features most correlated with the class label in the Elliptic++ dataset. Features related to transaction value (USD/BTC totals, means, and maxima) have the strongest positive correlations, indicating that illicit and licit transactions differ slightly in monetary patterns. Structural features such as transaction degree, size, and number of input/output addresses also contribute but with weaker effects.

REAL CATS:

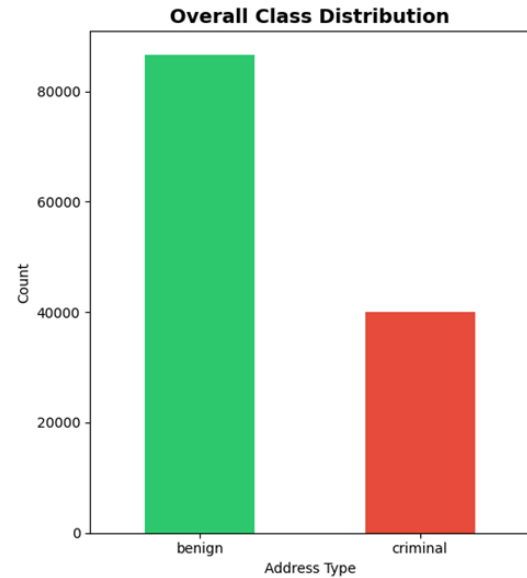
Data composition:

- Total addresses: 126,615 Bitcoin addresses
- Criminal Addresses: 40,032 (31.6%)
- Benign Addresses: 86,583 (68.4%)
- Class Balance: 1:2.16 ratio

Filtering: Removed zero-transaction addresses

Data quality:

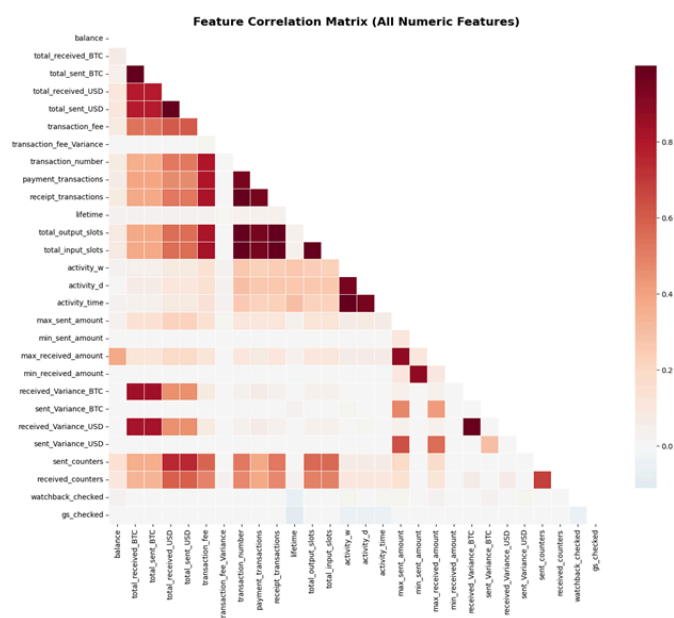
- Complete Features: 26 out of 37 features
- High Missing (>10%): 10 features (mostly transaction detail fields)



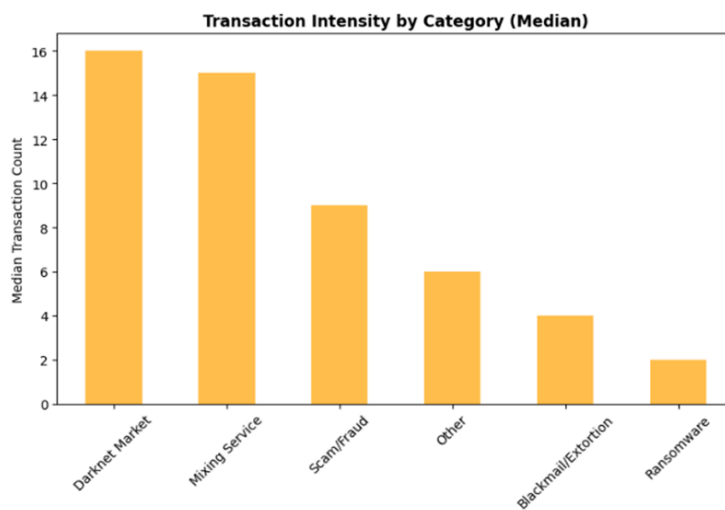
Metric	Criminal Median	Benign Median	Difference	Significance
Transaction Count	4.0	2.0	2.0× higher	Criminal wallets are more active
Total Received (USD)	\$2,433	\$1,591	53% higher	Criminal wallets receive higher volumes
Total Sent (USD)	\$2,426	\$1,049	131% higher	Criminal wallets send much more money

This table compares key behavioral metrics between criminal and benign wallets. **Criminal wallets are more active and handle larger transaction volumes**, especially in outgoing transfers. The large difference in total sent value suggests that criminal wallets tend to move funds more aggressively, making activity and volume-based features useful for distinguishing wallet behavior.

This figure shows the correlation matrix of all numeric features, highlighting pairs with near-perfect correlations. Several feature pairs (e.g., total BTC/USD received vs. sent, input vs. output slots, and closely related activity counters) capture almost identical information. These strong correlations indicate feature redundancy, which can negatively impact some models and inflate feature importance.



This figure shows the distribution of criminal address categories. Scams and ransomware are the most common types of illicit activity, accounting for the majority of criminal addresses and overall volume. In contrast, mixing services and darknet markets appear far less frequently, representing smaller and more specialized parts of the ecosystem.



This figure compares median transaction intensity across criminal categories. Darknet markets and mixing services show the highest transaction frequencies, indicating more structured and continuous operations that require steady liquidity. In contrast, scams, extortion, and ransomware have much lower transaction counts, reflecting more opportunistic behavior that relies on short-lived

or disposable wallets.